| INCH-POUND |

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10 April 1992
SUPERSEDING
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15 July 1980

MILITARY SPECIFICATION

HEATERS, WATER, LIQUID FUEL (OIL-FIRED, STORAGE TYPE: NOT EXCEEDING 280,000 BTU/H INPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical, light-oil fired, storage-type water heaters having thermal inputs not exceeding 280,000 British thermal units per hour (Btu/h) and storage capacities up to 100 gallons (gal) and constructed to Underwriters Laboratories, Inc. (UL) or American Society of Mechanical Engineers (ASME) requirements.

1.2 Classification. Tanks furnished on the water heaters shall be of the following types and sizes; burners shall be of the following styles (see Table I, 6.2, and 6.3):

Type IV - Glass-lined steel.
Type VII - Nickel-copper alloy (70 percent Nickel (Ni), 30 percent Copper (Cu)).
Type VIII - Copper.
Type IX - Stainless steel.
Type X - Copper-Nickel alloy (90 percent Cu, 10 percent Ni).
Type XI - Electroless nickel plated steel overcoated with fluorocarbon-based polymer.
Type XII - Steel overcoated with fluorocarbon-based polymer.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter|

AMSC N/A FSC 4520

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Size 30 - 30-gallon nominal storage capacity, with 0.5 gallons per hour (gph) minimum firing rate.
Size 30 - 30-gallon nominal storage capacity, with 0.75 gph minimum firing rate.
Size 40 - 40-gallon nominal storage capacity, with 0.85 gph minimum firing rate.
Size 50 - 50-gallon nominal storage capacity, with 1.20 gph minimum firing rate.
Size 70 - 70-gallon nominal storage capacity, with 1.35 gph minimum firing rate.
Size 85 - 85-gallon nominal storage capacity, with 1.45 gph minimum firing rate.
Size 100 - 100-gallon nominal storage capacity, with 1.45 gph minimum firing rate.

Style B - High pressure-air atomizing, gun type.
Style C - Low pressure-atomizing, gun type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-T-704 - Treatment and Painting of Material.
MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-C-26074 - Coatings, Electroless Nickel, Requirements for.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF ENERGY (DoE)
FEDERAL ENERGY ADMINISTRATION (FEA)

Title 10, Code of Federal Regulations (CFR), Chapter II - Part 430 and Amendments - Energy Conservation Program for Appliances.

(Application for copies of DoE/FEA documents should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
B1.20.1 - Pipe Threads, General Purpose (Inch).
B1.20.3 - Dryseal Pipe Threads (Inch).
Z21.22 - Relief Valves and Automatic Gas Shutoff Devices for Hot Water Supply Systems.

(Application for copies should be addressed to the American National Standards Institute, 11 W. 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion-Welded Unfired Pressure Vessels.
A 424 - Steel Sheets for Porcelain Enameling.
A 527 - Steel Sheet, Zinc Coated (Galvanized) by the Hot Dip Process Lock-Forming Quality.
B 107 - Magnesium Alloy Extruded Bars, Rods, Shapes, Tubes, and Wire.
B 122 - Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate Sheet, Strip and Rolled Bar.
B 127 - Nickel-Copper Alloy (UNS N04400) Plate, Sheet and Strip.

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B 152 - Copper Sheet, Strip, Plate, and Rolled Bar.
D 396 - Fuel Oils.
E 380 - Practice for Use of the International Systems of Units (SI).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code.
Section IV, Heating Boilers.
Section VIII, Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS-1 - General Standards for Industrial Control and Systems.
ICS-2 - Standards for Industrial Control Devices, Controllers and Assemblies.
ICS-6 - Enclosures for Industrial Controls and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL296 - Oil Burners.
UL378 - Draft Equipment.
UL732 - Oil Fired Storage Tank Water Heaters.
UL873 - Temperature-Indicating and Regulating Equipment.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The water heaters shall consist of an insulated vertical, steel, water storage tank, a light oil-fired combustion system not exceeding 280,000 Btu/h input capacity, and the operating and safety controls specified herein for completely automatic operation of the heaters for use in applications requiring potable water (see 6.1).

3.2 Standard commercial product. The water heaters shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the water heaters being furnished. Standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM E380.

3.3 First article. When specified (see 6.2), the contractor shall furnish a complete heater of each type, size, and style for first article inspection and approval (see 4.2.1 and 6.5).

3.4 Codes and standards. The water heater shall comply with the requirements of UL732 or ASME Boiler and Pressure Vessel Code, Section IV. The burner shall comply with UL296.

3.5 Energy conservation. Water heaters with thermal input of 105,000 Btu/h or less shall comply with DoE regulations, Title 10, CFR, Chapter II, Part 430 - Energy Conservation Program for Appliances.

3.6 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.6.1 Steel. Steel for fabrication of the storage tank shall be made of commercial quality sheet or plate, chemically constituted and suitably processed to adapt the steel for application of glass (porcelain enamel) linings. Steel for tanks with glass (porcelain enamel) lining may also conform to ASTM A424, be suitable for exposure to fire and products of combustion to which the tank will be exposed or may meet the material requirements of ASME Boiler and Pressure Code, Section IV, Heating Boilers. Steel shall be suitable for application of the lining material.

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3.6.2 Zinc-coated steel. Zinc-coated steel for fabrication of flue pipe accessories (see 3.18) shall conform to ASTM A527 with G60 coating having a thickness not less than 0.0276 inch (24 gage galvanized sheet gage).

3.7 Interchangeability. All heaters of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Design. The units shall be designed to permit easy accessibility for maintenance, cleaning, and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. Material in contact with potable water shall be nontoxic, and shall not impart odor, color, or taste to the contained water. Heaters shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation. Heaters shall be designed for standard clearance for ceilings, walls, floors, and flue pipe specified in UL732 without exceeding the applicable maximum temperature rises specified therein. The water heater shall be suitable for installation in an alcove or a closet as specified (see 6.2). The water heater shall be suitable for installation on noncombustible floor or combustible floors as specified (see 6.2). When the water heaters are required for installation on combustible floors, installation instructions shall be provided on the unit for such application. Heaters shall be designed to meet the capacity and performance requirements as follows:

(a) For heaters 105,000 Btu/h or less firing 138,400 Btu/gal oil (No. 1 oil) recovery capacity shall be based on DoE regulations, CFR Title 10, Chapter II, Part 430 - Energy Conservation Program for Appliances, as sole determination (see 6.6).

(b) For heaters firing 140,000 Btu/gal oil (No. 2 oil) see Table I.

TABLE I. Capacity and performance based on 140,000 Btu/gal of oil.

Efficiency percent	Nominal Storage capacity (US gallons)	Permissible deviation from nominal capacity (percent)	Firing rate min gph	Recovery capacity, min gph		
				60 deg rise	80 deg rise	100 deg rise
75	30	+/-5	0.5	106	79	63
	30	+/-5	0.75	159	119	95
70	40	+/-10	0.85	168	126	101
	50	+/-10	1.20	237	178	142
	70	+/-10	1.35	267	200	160
	85	+/-10	1.45	287	215	172
	100	+/-10	1.45	287	215	172

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3.8.1 Threads. Screw threads for all thread-securing or -attaching devices and threaded parts shall be as follows:

(a) Machine threads shall be the standard national coarse or fine thread series in accordance with ANSI B1.1.

(b) Pipe threads shall be the standard National Taper Pipe Thread (NPT) series in accordance with ANSI B1.20.1, except 1/8-inch to 3-inch pipe thread may be Dryseal American NPT in accordance with ANSI B1.20.3.

3.9 Performance. Performance and capacity requirements specified herein shall be based on firing light fuel oil having a heat content of 138,500 Btu/gal for input 105,000 Btu/h (No. 1 oil) or less, water heaters, or 140,000 Btu/gal (No. 2 oil) for all water heaters as specified (see 6.2). Operation of the heaters shall be under standard atmospheric conditions (60 degrees Fahrenheit (^{o}F)) and 14.7 pounds per square inch (psi) atmosphere with an inlet water supply temperature of 70°F +/-5°F and nominal specific heat of water of 8.25 Btu/gal at 120°F.

3.9.1 Combustion. Combustion shall be stable and complete at all firing rates of the burners. Observed smoke at all firing rates under steady-state test conditions shall not exceed the limits indicated by No. 1 on the Shell-Bacharach Scale with the Model RDC Smokemeter.

3.9.2 Water temperature control. When the heater is tested in accordance with 4.4.1.6, the water temperature shall not exceed 200°F.

3.9.3 Water temperature limit. When the heater is tested in accordance with 4.4.1.7, the water temperature shall not exceed 210°F.

3.9.4 Temperature rise for heater components. When the water heater is operated at rated output, no part shall attain a temperature sufficient to damage protection against corrosion, to adversely affect operation of safety controls, to impair the value of required thermal or electrical insulation, or to cause damage to material or to parts which will render the heater unsafe for use. The temperature rise at specific points on and adjacent to the heater shall not exceed the maximum allowable temperature rises specified in UL732.

3.9.5 Recovery efficiency.

3.9.5.1 For heaters with oil having heat content of 138,500 Btu/gal. The recovery efficiency for heaters of 105,000 Btu/h or less shall be in accordance with DoE/FEA regulations, CFR, Title 10, Chapter II, Part 430, and shall be the sole determination.

3.9.5.2 For heaters with oil having heat content of 140,000 Btu/gal. The minimum recovery efficiency for heaters with thermal input of 105,000 Btu/h or less shall be 75 percent and for heaters with thermal input greater than 105,000 Btu/h shall be 70 percent. The recovery efficiency shall be calculated and defined by the dimensionless quantity of:

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$$E_r = \frac{(k)\ (V)\ (\text{Delta T})\ 100}{(Q_r)\ (H)}$$

k = 8.25 Btu/gal °F nominal specific heat of water at 120°F.
V = Tank recovery capacity in gallons as determined by weighing in accordance with 4.4.1.1.2.
Delta T = The difference between the average initial and final mean tank temperatures as determined in 4.4.1.1.2.
Q_r = Total flow of fuel expressed in gallons.
H = Higher heating value for fuel in Btu/gal.
E_r = Recovery efficiency percent.

3.10 Tank. The water storage tank shall be of the vertical type and shall be designed for the applicable working pressure and test pressures specified in Table II. Tanks shall also withstand the proof test pressure specified in Table II without damage or failure of the lining due to such defects as crazing, cracking, or spalling. The tank may be constructed in accordance with ASME Boiler and Pressure Vessel Code, Section IV, Pressure Vessels. When the cold water inlet is located in the top of the tank, a dip tube shall be provided. An antisiphoning hole shall be so located in the dip tube that, when the tube is installed, the hole will be no more than 6 inches from the top of the tank. Nonmetallic dip tubes shall be capable of withstanding a temperature of 230°F. Type IV glass-lined (porcelain-enameled) tanks shall be equipped with anodes as specified in 3.11.

TABLE II. Design pressures.

Requirement	Standard pressures (psig) (min)
Working pressure	150
Proof test pressure	150
Hydrostatic test pressure	300

3.10.1 Tappings and fittings. Tappings and fittings for glass-lined tanks shall be carbon steel, copper silicon, copper-lined steel, brass, tin- or brass-plated steel or copper-plated steel and shall be furnished with insulating bushings with instructions on their use attached to the units. For copper, nickel-copper and stainless steel tanks, the tappings and fittings shall be copper silicon, stainless steel or the same material as the tanks. Tappings and fittings shall have tapered threads in accordance with ANSI B2.1. A separate tapping for a pressure relief valve shall be furnished in the tank and shall be not less than 3/4-inch pipe size and shall be on the top or within 6 inches of the top of the tank. The pipe size of the inlet and outlet tappings shall be not less than shown in Table III.

TABLE III. Inlet and outlet water tappings.

Size tank	Pipe size inches minimum
30, 40, 50	3/4 or 1-1/4
70, 85	1-1/4
100	1-1/2

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3.10.2 Tank linings. Linings for tank shall be the type(s) as specified herein. When specified (see 1.2 and 6.2), the invitation for bids shall indicate that any one of two or more linings will be acceptable. The manufacturer shall select from the specified alternatives the particular lining he will furnish and this lining shall be applied to all heaters of the same type furnished under the contract. Prior to application of linings, the interior surfaces of the tank, including welds, shall be prepared for coating by sand or grit blasting or by chemical treatment.

3.10.2.1 Type IV, glass-lined steel tank (porcelain enamel coating). The glass lining shall be applied to all interior surfaces that are exposed to hot water including handhole and handhole covers when applicable. The glass lining shall be fused to the steel base by firing at a temperature of not less than 1500°F. Type IV heaters shall conform to the following requirements:

(a) Thickness. The thickness of the glass lining shall be not less than 0.006 inch and not greater than 0.020 inch.

(b) Discontinuities. Discontinuities shall be not less than 0.025 square inch per square foot of internal surface, excluding edges, fittings and welded areas, after applying the glass lining. No single defect shall be greater than 0.125 inch in diameter.

(c) Edges and fittings. All edges and fittings shall be coated with glass lining. Where sharp edges cannot be avoided, a slight "burn-off" of the lining that does not extend more than 0.062 inch back from the sharp edge will be allowed. All fittings that are welded to the tank shall be coated with the lining on those areas that are exposed to the water except for threads and small areas immediately adjacent to them.

(d) Welds. Removal of the glass lining due to welding or mating of parts after glass lining shall not result in exposed base metal in excess of 0.025 square inch per gallon of storage capacity.

(e) Glass lining solubility. The weight loss of the glass lining shall not exceed 15 milligrams per square inch.

3.10.2.2 Type VII, nickel-copper alloy tank. Nickel-copper alloy shall conform to ASTM B127, alloy USN N04400 cold rolled and annealed.

3.10.2.3 Type VIII, copper tank. Copper shall conform to ASTM B152, not less than alloy USN C12500 cold rolled, temper H00.

3.10.2.4 Type IX, stainless steel tank. Stainless steel shall conform to ASTM A167, alloy S30400, S30403, S31603; No. 1 finish, or ASTM A240, S43035.

3.10.2.5 Type X, copper-nickel alloy tank. Copper alloy shall conform to ASTM B122, alloy USN No. C70600, temper 060.

3.10.2.6 Type XI, electroless nickel-plated steel overcoated with fluorocarbon based polymer. The steel tank shall be internally electroless nickel-coated as specified in MIL-C-26074, class 2, grade A, and shall withstand proof pressure of Table II herein without cracking. After complete fabrication, the tank shall be lined over the electroless nickel with three separate

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applications of polymerized fluorocarbon with individual curing cycles at a temperature of a minimum of 450°F and a maximum of 460°F. The thickness of the lining shall not be less than 0.006 inch. Discontinuities shall be not more than 0.025 square inch per square foot of internal surface. No single defect shall be greater than 0.125 inch in diameter. The weight loss due to solubility of the fluorocarbon polymerized lining shall not exceed 15 milligrams per square inch.

3.10.2.7 Type XII, steel overcoated with fluorocarbon based polymer. The steel tank, after complete fabrication, shall be lined with three separate applications of polymerized fluorocarbon with individual curing cycles at a temperature of a minimum of 450°F and a maximum of 460°F. The thickness of the lining shall not be less than 0.006 inch. Discontinuities shall be not more than 0.025 square inch per square foot of internal surface. No single defect shall be greater than 0.125 inch in diameter. The weight loss due to solubility of the fluorocarbon polymerized lining shall not exceed 15 milligrams per square inch.

3.10.3 Handhole. When specified (see 6.2), the tank shall be furnished with a handhole for inspection and cleaning. The handhole cover shall have the same lining as the tank and be gasketed or made secure so that it will not leak at the test pressure prescribed for the tank.

3.11 Cathodic protection. Unless otherwise specified (see 6.2), type IV heater shall be provided with magnesium anodes for cathodic protection. The anode shall be located so as to insure protection to all interior parts of the vessel. An electrical connection shall be established and maintained between the anode and the metal tank (heater) shell.

3.11.1 Magnesium anode. The magnesium anode shall be fabricated from material conforming to ASTM B107, composition USN No. N11311 with the following exceptions:

Material	Maximum percent
Copper	0.04
Nickel	0.001
Iron	0.002

Shapes of anodes shall be as applicable, and anodes shall be reinforced with steel wire not less than 0.091 inch in diameter. The weight of the anode in each tank shall be not less than 25 grams per square foot of surface within the tank shell.

3.11.2 Other anodes. The tank (heater) wall potential for any anode used shall not be more positive than -0.85 volt.

3.11.3 Caution notice. Heaters equipped with anodes shall carry a "CAUTION" notice with essentially the following message on the outside (see 3.23):

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"CAUTION"

"Hydrogen gas can be produced in a hot water system that has not been used for a long period of time (generally 2 weeks or more). Hydrogen gas is extremely flammable. To prevent the unlikely possibility of injury under these conditions, we recommend the hot water faucet be open for several minutes at a sink before you use any electrical appliance which is connected to the hot water system. If hydrogen is present, there will probably be an unusual sound, such as air escaping through the pipe as the hot water begins to flow. There should be no smoking or open flame near the faucet at the time it is open."

3.12 Flues. Each heater shall be designed with one or more passageways, which serve as flues to conduct the hot gases from the combustion chamber through the heater to the connection smoke pipe. The flue(s) shall be either the internal-pipe type or the external type that conducts the hot gases along the outside surface of the tank. The flue(s) shall be designed to assure maximum heat transfer from the hot gases to the water and shall be accessible for periodic cleaning. The flue(s) shall terminate in a flue collar designed to permit attachment of a sheet metal smoke pipe. The flue collar may be located for either top vertical discharge or rear horizontal discharge.

3.13 Insulation and jacket. The entire unit shall be insulated with fibrous glass, mineral wool, or equivalent insulation. The insulation shall be securely held in place with a steel jacket not less than 0.0239-inch (US revised standard gage No. 24) thick. Necessary openings and doors shall be provided for easy access to the combustion chamber and the flue and for removal of the burner. The insulation and jacket shall be adequate to insure performance specified in 3.9 and temperature limitations of UL732.

3.14 Combustion chamber. The combustion chamber shall be equipped with suitable supports for mounting the burner beneath the flue pipe. Provisions shall be made for mounting the burner through one side and to adequately seal the combustion space so that flames do not impinge on the tank insulation. The walls and floor of the combustion chamber shall be lined with lightweight firebrick; fireclay plastic refractory; or molded, interlocked, refractory shapes, rigidly supported in place.

3.15 Oil burners. The oil burners shall be in accordance with UL296. The minimum firing rate for the burners shall be as specified in 1.2 under size when firing No. 1 or No. 2 oil per ASTM D396 without exceeding No. 1 smoke limit as indicated on the Shell-Bacharach Scale with Model RDC Smokemeter. Unless otherwise specified (see 6.2), the burners shall operate on 115 volts, single phase, 60 hertz. When specified (see 6.2), a motor starter conforming to NEMA ICS-1 and NEMA ICS-2 shall be provided as a separate item with a NEMA 1 enclosure conforming to NEMA ICS-6 with overload protection.

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3.16 Controls. Unless otherwise specified (see 6.2), controls shall operate on 115 volts, single phase, 60 hertz.

3.16.1 Thermostat. The water heater shall be equipped with a thermostat conforming to UL873. The thermostat shall be of the adjustable immersion type, having a well installed at the proper point in the heater tank, and actuated by water temperature to start and stop the burner. The thermostat shall be designed to maintain control within +/-10°F of the thermostat setting, shall encompass a range from 100°F to at least 160°F, and will not allow the water to attain a temperature of more than 200°F. Temperature settings shall be easily accessible from the front of the heater and adjustable by hand, but shall be so protected that they cannot be accidentally disturbed. The thermostat shall be set at its lowest setting for shipment with warning instructions supplied about burn hazard potential.

3.16.2 Limit control cutout. The water heater shall be provided with an automatic fuel shutoff system. The shutoff system shall be of the manual reset type and shall function in accordance with UL732 to interrupt the fuel supply to the heater before the water temperature exceeds 210°F and after the water temperature is higher than that maintained by the temperature regulating control of 3.16.1. The limit controller shall be similar to the thermostat; shall have no operating parts in common with the temperature regulating device but may have a common mounting bracket or a common enclosure. The temperature limiting device shall be located so that the temperature sensitive element controls the temperature of the water within the top 6 inches of the tank.

3.16.3 Combustion safeguard. The burner (see 3.15) for the water heater shall be equipped with combustion safeguard (primary control) conforming to UL296. The combustion safeguard shall affect burner shutdown in event of ignition failure, startup, or flame failure during the firing cycle.

3.16.4 Wiring. Wiring shall be in accordance with NFPA No. 70. Provisions shall be made to ground all electrical equipment. A wiring harness shall be provided between the cod cell relay and thermostat to assure correct factory installed controls.

3.17 Combination pressure-temperature relief valve. The water heater shall be furnished with a combination pressure-temperature relief valve conforming to ANSI Z21.22. The rated discharge capacity of the pressure and temperature relief elements shall be not less than the hourly Btu input to the heater for which the valve is furnished. Discharge capacities shall be established in accordance with ANSI Z21.22. The set pressure for the pressure relief element shall be in accordance with ANSI Z21.22. The temperature elements shall be set to prevent the water temperature in the heater from exceeding 210°F. The valve shall be not less than 3/4-inch pipe size and installed in a manner to insure immersion of the thermostatic element within the upper 6 inches of the storage tank and shall be in a separate tapping other than the tappings for the water connections (see 3.10.1).

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3.18 Accessories. When specified (see 6.2), one or more of the following accessories shall be furnished with each heater:

(a) Downdraft cap	1 each
(b) Storm collar	1 each
(c) Elbow 90°, adjustable for smoke pipe	2 each
(d) Smoke pipe, chimney or vent connector for heater	sections as specified (see 6.2)
(e) Draft regulator, barometric for smoke pipe	1 each
(f) Insulating couplings for inlet and outlet water connections	1 set
(g) Electrodes, ignition for fuel oil burner	2 each
(h) Electrode lead, ignition for fuel oil burner	2 each

3.18.1 Downdraft cap. The downdraft cap for water heaters shall be fabricated of zinc-coated steel and shall be designed to fit securely over the top of the smoke pipe specified in 3.18.4. The cap shall be of the vacuum type designed to eliminate downdrafts due to wind action across the smoke pipe and to increase the draft when a negative pressure develops within the building.

3.18.2 Storm collar. The storm collar for water heaters shall be fabricated of zinc-coated steel and shall be designed to fit securely around the smoke pipe. The collar shall be designed to eliminate water running down between the flue pipe and the flashing. The storm collar shall be designed to be adjustable, to accommodate a flat roof, or a sloped roof with slope as specified (see 6.2).

3.18.3 Elbow 90 degrees adjustable. The elbow shall be fabricated of zinc-coated steel and shall fit securely over the smoke pipe specified in 3.18.4. The elbow shall be manually adjustable from 90° bend to a straight pipe section. The seaming shall be so finished as to provide a uniform inside surface at the crimped end. The method of crimping shall prevent separation at the adjustable joint and minimize leakage of the flue gages.

3.18.4 Smoke pipe, chimney, or vent connector. The smoke pipe shall consist of 24-inch long sections, with number of sections as specified (see 6.2). Each section shall be fabricated from one piece of zinc-coated steel. The longitudinal lock seam shall be of the snap locking design. If the flue collar on the heater is other than the size of the flue pipe furnished, a transition piece shall be furnished.

3.18.5 Draft regulator. The draft regulator shall be of the barometric type, designed for installation in the chimney or vent connector at the outlet of the heater and screw adjustable to encompass a range of settings from 0.02 to 0.08 inch of water. The draft regulator shall meet the requirements of UL378.

3.19 Treatment and painting. Unless otherwise specified (see 6.2), the heater shall be treated and painted in accordance with the manufacturer's standard commercial practice. When specified (see 6.2), the heater shall be treated and painted in accordance with MIL-T-704, type A or C, with the finish coat color in accordance with FED-STD-595, as specified (see 6.2).

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3.20 Electromagnetic interference suppression. When specified (see 6.2), the equipment shall be designed and equipped for electromagnetic interference suppression in accordance with class C3, group II of MIL-STD-461.

3.21 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.

(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.22 Identification marking. Unless otherwise specified (see 6.2), the heater shall be marked for identification in accordance with manufacturer's standard practice. When specified (see 6.2), the contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and subassembly and parts as directed by the Government inspector. Brass screws, rivets, or bolts not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "WATER HEATER, OIL-FIRED STORAGE TYPE."

3.23 Instruction plate. When any special or important procedure is to be followed in operating, installing, or servicing the equipment, the unit shall be equipped with an instruction plate suitably located, describing such procedure. Plates shall be of a material which will last and remain legible for the life of the equipment and shall be securely affixed to the heater in accordance with the manufacturer's standard practice.

3.24 Workmanship. The quality of workmanship shall be such as to produce heaters that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the unit.

3.24.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.24.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.24.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of uniform shape and shall be concentric with the rivet. Rivet heads shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

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3.24.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.24.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Material inspection. The contractor is responsible for ensuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.1.3 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one complete water heater when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the

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requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3.2, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination.

4.3.1 In-process examination. Heaters furnished under this specification shall be examined during production in accordance with the manufacturer's established quality control procedures, which shall assure compliance with all requirements of this specification that can be checked visually, and with those elements of the manufacturer's drawings or specifications normally verified during the inspection process. In-process examination shall apply to all areas of contract performance including the receipt and identification of material, fabrication of components, finishing of parts, and final assembly of the heaters. Quality control procedures shall be subject to surveillance by the procuring agency or by a designated representative thereof. Failure of the manufacturer to demonstrate that adequate quality control was applied to lots submitted for acceptance by the Government may be cause for rejection of the lots.

4.3.2 End-item examination. Each water heater shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects detrimental to specified performance shall constitute cause for rejection.

4.4 Tests.

4.4.1 First article tests. The first article sample, when required, shall be tested. Tests shall be conducted as outlined and as specified herein.

4.4.1.1 Recovered efficiency.

4.4.1.1.1 For units firing 138,500 Btu/gal oil with input rates of 103,875 Btu/h and under. Acceptable evidence of the recovery efficiency of units under 103,875 Btu/h shall be the table prescribed by the DoE/FEA regulations, Title 10, CFR, Chapter II, Part 430.

4.4.1.1.2 For units firing 140,000 Btu/gal oil for all input rates. The water heater shall be installed by the manufacturer with thermostats in the cold water inlet line, the hot water outlet line, and a line from the drain connection. The line from the drain connections shall be arranged to discharge into a container in order to weigh the water. Provision shall be made for shutting off the above lines. The oil rate shall be metered to the burner and be adjusted and maintained at normal pressure. Before starting any test, the setting of the thermostat shall be obtained by starting the water in the system

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at 70 +/-5°F, and noting the maximum temperature of the water drawn from the hot water outlet immediately after the thermostat shuts off the oil supply. The temperature shall be 160 +/-5°F. When the thermostat shuts off the oil supply, the inlet water and main burner valves shall be closed and the water drained. After the system is emptied, the drain valve shall be closed, the inlet water valve opened, and the system filled at once with water at 70 +/-5°F. The burner meter reading shall be noted and the heater shall again be put into operation. When the thermostat shuts off the oil supply, the inlet water valve and oil valve shall be closed immediately, the burner meter reading taken, and the water in the storage vessel drained. The heat content of the water shall be determined by allowing the water to flow into a container and reading its temperature at 10-pound intervals, noting the total weight as well. The high heat content of the oil shall be as in 3.9. Efficiencies shall be calculated by means of the equation in 3.9.5.2. Failure to meet the efficiency specified in Table I shall constitute failure of this test.

4.4.1.2 Storage capacity. The storage capacity shall be determined by weighing the system when dry and empty and reweighing it when full or by filling the system with water, the weight of which has been predetermined. The capacity shall then be computed in gallons and compared with the manufacturer's rated capacity. Failure to have a capacity within the limits of the values for the size under consideration of Table I shall be cause for rejection of the heater.

4.4.1.3 Relief valve. The relief valve shall be tested to verify conformance with the pressure and temperature relieving requirements of 3.17. During the test run, the thermostatic control for the water temperature shall be disconnected or otherwise rendered inoperative. The heater shall be equipped with pressure gages and thermometer and shall be operated at full rated capacity. The relieving capacity of the temperature element shall be tested first to verify that the relieving capacity is sufficient to prevent the water temperature in the tank from exceeding 210°F. The temperature element shall then be blanked off or otherwise rendered inoperative and the pressure element tested to verify that the relieving capacity is sufficient to prevent the pressure in the tank from exceeding the set pressure of the valve by more than 10 percent. The certification symbol of the American Gas Association Laboratories or a nationally recognized testing laboratory that maintains periodic inspection of production, will be acceptable evidence of meeting this test provided the relief valve is plainly stamped, cast, or etched on the device or on a permanent nameplate securely affixed to the device with a set pressure and temperature and with a maximum heater input at least equal to or greater than the maximum heaters supplied under the contract. Nonconformance to 3.17 shall constitute failure of this test.

4.4.1.4 Cathodic protection test. For type IV heaters, the heater shall be filled with water having a specific conductance in the range of 400 to 600 micro-mhos. The tank wall potential shall be measured in reference to a standard copper sulfate cell located inside the tank adjacent to, but not in contact with, the tank shell and at a maximum distance from the anode or anodes. Measure the tank wall potential with a portable potentiometer or high-resistance voltmeter (see 3.11). Upon approval of the contracting officer, the manufacturer may furnish a certificate of compliance that the heaters meet the requirements of the wall potential in 3.11.2 with the data upon which certification is based. Failure to meet the requirements of 3.11.2 shall be cause for rejection of the heater.

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4.4.1.5 Burner operation and smoke test. When the heater is fired at rated input and that the stack loss is not more than 25 percent and operated until steady state, conditions of draft, fuel-input rate, and flue-gas temperature have been established, the smoke in the flue gages shall not exceed that indicated by No. 1 on the Shell-Bacharach Scale with the Model RDC Smokemeter. Failure to meet the requirements shall be cause for rejection of the heater.

4.4.1.6 Water-temperature control test. The heater is to be filled with water at room temperature +/-5°F. The temperature control is to be adjusted to its maximum allowable setting. A pressure-relief device followed by a quick-acting valve is to be installed on the outlet connection of the storage vessel. A flow restriction, calibrated to permit a flow of 5 gallons per minute (gpm), is to be connected to the outlet of the valve. A water-pressure regulator is to be located in the water supply line to the heater and adjusted so that at full flow, the pressure at the inlet connection to the heater will be maintained at the value required to deliver a steady flow of 5 gpm during test-draw periods. During the test, the inlet-water temperature is to be maintained at 70 +/-5°F. The heater is to be fired at rated high-fire input with the inlet-water valve opened and the outlet-water valve closed, until the temperature control functions to reduce the fuel supply to the burner to a minimum. Water is to be then drawn immediately at the rate of 5 gpm until the control functions to turn on the fuel supply to the burner, and the maximum outlet-water temperature is recorded. This operation is to be repeated until a constant or continually receding outlet-water temperature is attained. The maximum outlet-water temperature is not to exceed 200°F as indicated in 3.16.1 at any time during the test. Water temperatures exceeding 200°F will constitute failure of this test and shall be cause for rejection of the heater.

4.4.1.7 Limit-control cutout test. With the water heater installed as in 4.4.1.6, the temperature regulating control is to be made inoperative. The heater is to be fired at rated high-fire input with the inlet-water valve opened and the outlet-water valve closed until the limit control functions. The water temperature shall not exceed the requirements specified in 3.16.2. Water temperatures exceeding the requirements of 3.16.2 or failure to shut off the burner shall be cause for rejection of the heater.

4.4.1.8 Glass lining tests. Type IV glass (porcelain enamel) linings shall be inspected and tested as follows for coverage, thickness, and solubility. Upon approval of the contracting officer, the manufacturer may furnish a certification of compliance as part of the first article. The certification shall provide data necessary to evaluate compliance with the coverage, thickness, and solubility requirements of 3.10.2.1. Nonconformance with 3.10.2.1 shall constitute failure of this test.

4.4.1.8.1 Lining coverage. A glass lined tank that has not been subjected to test pressure greater than 150 pounds per square inch gage (psig) shall be cut into four or more segments using a band saw, bayonet saw, or other suitable mechanical cutting device. Each segment shall be visually inspected for the presence of areas of exposed metals except that cracked and spalled areas of lining caused by the cutting operation shall not be considered. Exposed areas of metal shall be measured with a low powered microscope with a calibrated eyepiece. Nonconformance to 3.10.2.1.b and 3.10.2.1.c shall constitute failure of this test.

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4.4.1.8.2 Thickness of glass lining. Thickness measurements shall be taken at not less than 10 representative points on the shell and heads of the tank using a calibrated magnetic or similar thickness gauge. Nonconformance to 3.10.2.1.a shall constitute failure of this test.

4.4.1.8.3 Solubility of glass lining.

(a) Apparatus. The apparatus illustrated in figure 1 shall be used for determining the solubility of the glass lining. The cylinder assembly shall be constructed of 16 gage type 304 stainless steel; the ends shall be ground so as to conform with the curvature of the tank to be tested. The gaskets shall be made of neoprene rubber. The cell shall be heated with a small gas burner or a small electric heater placed immediately below the cylinder assembly midway between the ends.

(b) Test specimens. The test specimens for this test shall consist of four 3-1/2 inch by 3-1/2 inch sections cut from the outer wall (not the center flue) of the tank approximately midway between the ends. These specimens may be taken from the center segments cut in 4.4.1.8.2.

(c) Preparation of specimens. The specimens shall be buffed with a rubbing stone or suitable abrasive paper around the entire perimeter of the panel to completely remove rough edges and glass lining fragments. The specimens shall then be scrubbed on both sides using a nylon brush and a mildly abrasive detergent cleaner powder, rinsed with distilled water, dried for 1 hour in a drying oven at 225°F (107° Celsius) and placed in a desiccator while hot.

(d) Initial weight measurements. The specimens after cooling to room temperature, shall be weighed to the closest 1/10 milligram.

(e) Calibration of test cell. Because of small variations in cell dimensions and configuration, each test cell must be calibrated before it is used. The procedure is as follows:

(1) Assemble the cell as it is to be operated.
(2) Fill with water to just below where the condenser tube is welded into the cylinder (see figure 1).
(3) Adjust the input of the heat source to give a slow, rolling boil. If water rises in the condenser, remove small amounts of water until the cell will operate without surging.
(4) Remove from the heat and check to see if water level is completely covering the panels.
(5) Cool to room temperature, then measure the volume of water contained in the cell.
(6) Record this volume in the cell and use this amount of test solution in all subsequent tests.

(f) Test solution. The test solution used for each exposure cycle shall consist of 400 milligrams of reagent grade sodium bicarbonate dissolved in 1 liter of distilled water.

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(g) Test procedure.

(1) Assemble the test cell using weighed specimens from the same tank on each end of the test cell.
(2) Pour the correct volume of test solution into the cell.
(3) Adjust the input of the heater to give a consistent slow, rolling boil.
(4) After 18 hours boil time, dismantle the cell and discard the used test solution.
(5) Clean the specimens using a soft cloth wetted with a solution of 1 percent trisodium phosphate to remove loosely adhering deposits, wipe dry and store in desiccator between solution treatment.
(6) After 8 cycles of 18 hours each, clean specimens as above, rinse with distilled water and dry at 225°F (107° Celsius) for 1 hour. Place the specimens in a desiccator while hot and after cooling to room temperature weigh to the closest 1/10 milligram.

(h) Calculation of results. The exposed area of each specimen shall be calculated from the average diameter of the test spot, measured to the closed 0.015 inch (3.96mm) taken in three different locations. The solubility of the lining shall be calculated by dividing the weight loss in milligrams by the exposed area in square inches.

(i) Interpretation of results. Nonconformance with 3.10.2.1.e shall constitute failure of this test.

4.4.1.9 Electroless nickel-plated steel overcoated with fluorocarbon-based polymerized lining test. Type XI electroless nickel-plated steel overcoated with fluorocarbon-base polymer linings shall be inspected and tested as follows for plating thickness, plating adhesion, lining coverage, lining thickness and lining solubility. Upon approval of the contracting officer, the manufacturer may furnish a certification of compliance as part of the first article. The certification shall provide data necessary to evaluate compliance with plating and lining requirements in 3.10.2.6. Nonconformance with 3.10.2.6 shall constitute failure of this test.

(a) Compliance with the plating thickness requirement shall be determined in accordance with MIL-C-26074.
(b) Compliance with the plating adhesion requirement shall be determined in accordance with MIL-C-26074 and by the hydrostatic proof test in 4.4.2.1.
(c) Compliance with the lining coverage requirement shall be determined in accordance with 4.4.1.8.1 except that nonconformance with 3.10.2.6 shall constitute failure of the test.
(d) Compliance with the lining thickness requirement shall be determined in accordance with 4.4.1.8.2 except that nonconformance with 3.10.2.6 shall constitute failure of the test.
(e) Compliance with the lining solubility requirement shall be determined in accordance with 4.4.1.8.3 except that nonconformance with 3.10.2.6 shall constitute failure of the test.

4.4.1.10 Measurement of electromagnetic suppression. To determine conformance to the electromagnetic interference suppression of 3.20, electromagnetic emissions shall be measured in accordance with MIL-STD-462.

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4.4.2 Quality conformance tests.

4.4.2.1 In-process test. Each water heater shall be hydrostatically proof tested at the test pressure required in Table II and in accordance with the manufacturer's standard procedure. The symbol of the ASME Boiler and Pressure Vessel Code, Section VIII may be accepted as evidence of this test.

4.4.2.2 End item test. Each water heater shall be filled with water and the oil burner fired and controls shall be operated a sufficient period of time to demonstrate the heat input, smoke requirements and operation of the thermostat and limit controls and that the units are functioning as specified herein. After completion of tests, the unit shall be drained and serviced for shipment. Failure of any unit to pass this test shall be cause for rejection.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2), the heaters shall be preserved, packed, and marked in accordance with MIL-B-3180. The level of preservative and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Heaters covered by this specification are intended primarily for use in military residences and similar facilities. The heaters may be used in galleys, mess halls, showers, laundries, or shops where thermal input up to 280,000 Btu/h will satisfy the demand.

(a) Two or more heaters may be piped in series when demand load exceeds the output of a single heater.
(b) Where there is a continuous demand for hot water an indirect (coil type) water heater should be used.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, and style required (see 1.2 and 3.10.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).
(e) When first article is required for inspection and approval (see 3.3, 4.2.1, and 6.5).
(f) When water heater shall be suitable for installation in an alcove or closet (see 3.8).
(g) When water heater shall be suitable for installation on noncombustible floor or combustible floor (see 3.8).
(h) Whether heat content of fuel is 138,500 Btu/h or 140,000 Btu/h (see 3.9).
(i) When any one of two or more linings will be acceptable in the invitation for bids (see 3.10.2).

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(j) When handholes are required (see 3.10.3).
(k) If different number, type, and chemical composition of material of anode(s) are to be used (see 3.11).
(l) When electrical characteristics are different (see 3.15).
(m) When motor starter is required (see 3.15).
(n) When voltage to controls shall be different (see 3.16).
(o) Accessories required (see 3.18).
(p) When storm collar for flat roof, or sloped roof with a specified slope is required (see 3.18.2).
(q) Smoke pipe, chimney, or vent connector required with number of sections (see 3.18.4).
(r) When treatment and painting shall be different (see 3.19).
(s) When treatment and painting is required to be in accordance with MIL-T-704 (see 3.19).
(t) Color of finish coat required (see 3.19).
(u) When electromagnetic interference suppression is required (see 3.20).
(v) When fungus treatment is required (see 3.21).
(w) When identification marking is to be different (see 3.22).
(x) When contracting officer will furnish to the Government inspector the required identification plate (see 3.22).
(y) If preservation, packaging, packing, and marking is other than specified (see 5.1).
(z) Level of preservation and level of packing required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD form 1423.

Reference Paragraph	DID Number	DID Title
6.3.1.1 and 6.3.1.2	DI-MISC-80678	Certification Data/Report
6.3.1.3	DI-TMSS-80527	Commercial Manuals

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the water heaters and burners he proposes to furnish under this specification meet the applicable requirements of UL732, UL296, and ASME Boiler and Pressure Vessel Code, Section IV.

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6.3.1.1 ASME Boiler and Pressure Vessel Code, Section IV. Acceptable evidence of meeting the requirements of ASME should be the presence of the ASME Official Code HLW - symbol stamped or marked on the water heater and the furnishing of a copy of the applicable ASME Required Manufacturer's Master Data Report.

6.3.1.2 UL requirements. Acceptable evidence of meeting the requirements of UL732 and UL296 should be the certification symbol, or listing mark on the products, or a certified test report from a recognized independent testing laboratory indicating the water heaters have been tested and conform to UL requirements.

6.3.1.3 Technical manuals. The manufacturer should provide three commercial technical manuals normally prepared and supplied with the water heater.

6.3.1.4 Measurement of electromagnetic suppression. When suppressed to conform to the electromagnetic interference requirements of 3.20, the manufacturer may upon approval of the contracting officer furnish a certification in lieu of the test, that the heaters meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

6.4 Definitive specification part number. The specification part number is a definitive part number which corresponds to the types, sizes, firing rate, and burner style of the water heaters covered by this specification and defines the requirements of the options represented under this specification. The specification number, and the type and size code number are combined to form the definitive specification part number.

6.4.1 Cataloging data. For cataloging purposes, part numbers for the heaters are assigned as follows:

M16451-04-35-B

Specification part number
Type and number (see 6.4.2)
Size and minimum firing rate (see 6.4.3)
Style burner (see 6.4.4)

6.4.2 Type. The type of the heater (see 1.2) is identified by a two-digit number (see Table IV).

TABLE IV. Code number to type.

Tank type	Code number
IV	04
VII	07
VIII	08
IX	09
X	10
XI	11

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6.4.3 Size and minimum firing rate. The size and minimum firing rate of the heater (see 1.2 and Table I of 3.8) are identified by a two-digit number (see Table V).

TABLE V. Code number to size and minimum firing rate.

Tank size US gallons	Firing rate gph	Code number
30	0.5	35
30	0.75	37
40	0.85	48
50	1.20	52
70	1.35	73
85	1.45	84
100	1.45	14

6.4.4 Style of burner. The style of burner for the heater (see 1.2) is identified by a single letter (see Table VI).

TABLE VI. Code letter to style of burner.

Burner style	Code letter
B (high)	B
C (low)	C

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Capacity and performance for units 105,000 Btu/h or less. The water heaters should have the capacity and performance as shown in Table VII.

TABLE VII. Capacity and performance.

Nominal Storage capacity (US gallons)	Permissible deviation from nominal capacity (percent)	Firing rate min gph	Recovery capacity, min (75 percent efficiency based on 138,500 Btu per gallon of oil) 60 deg rise	80 deg rise	100 deg rise
30	+/-5	0.5	105	78	63
30	+/-5	0.75	157	118	95

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6.7 Subject term (key word) listing.

Water heaters
Oil-fired
Liquid fuel
Storage type

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

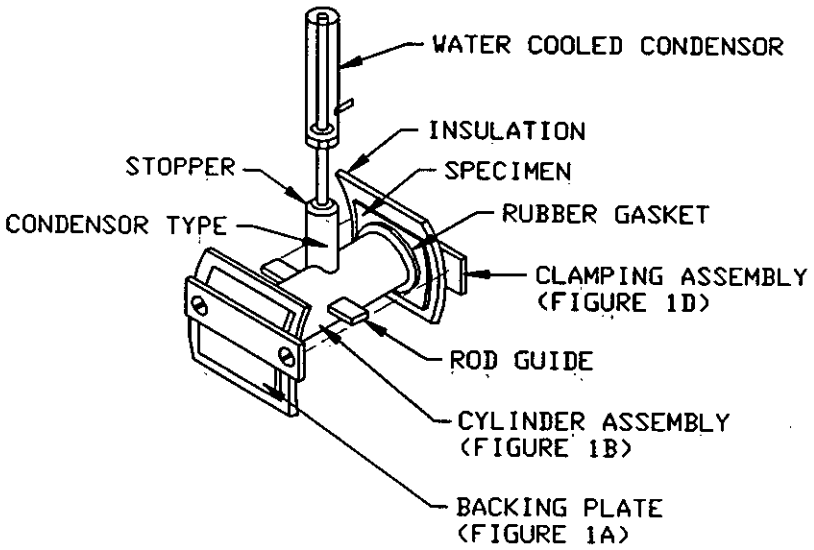
Preparing activity:
Navy - YD

(Project 4520-0338)

Review activities:
Army - CE
Air Force - 84
DLA - CS

User activity:
Navy - MC

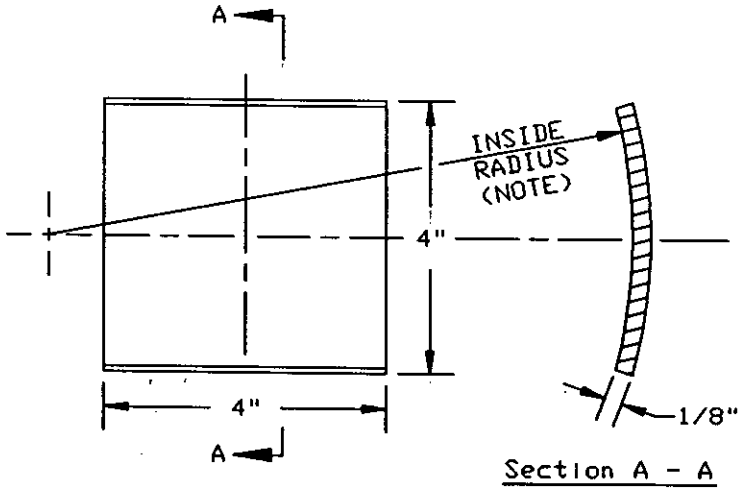
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Note:

1. Material: type 304 stainless steel throughout.

FIGURE 1. Test cell solubility of enamel.
(See also figures 1A to 1D)



Note:

1. The radius shall be the same as the inside radius of the tank from which specimens are cut.

FIGURE 1A. Backing plate. (2 required)

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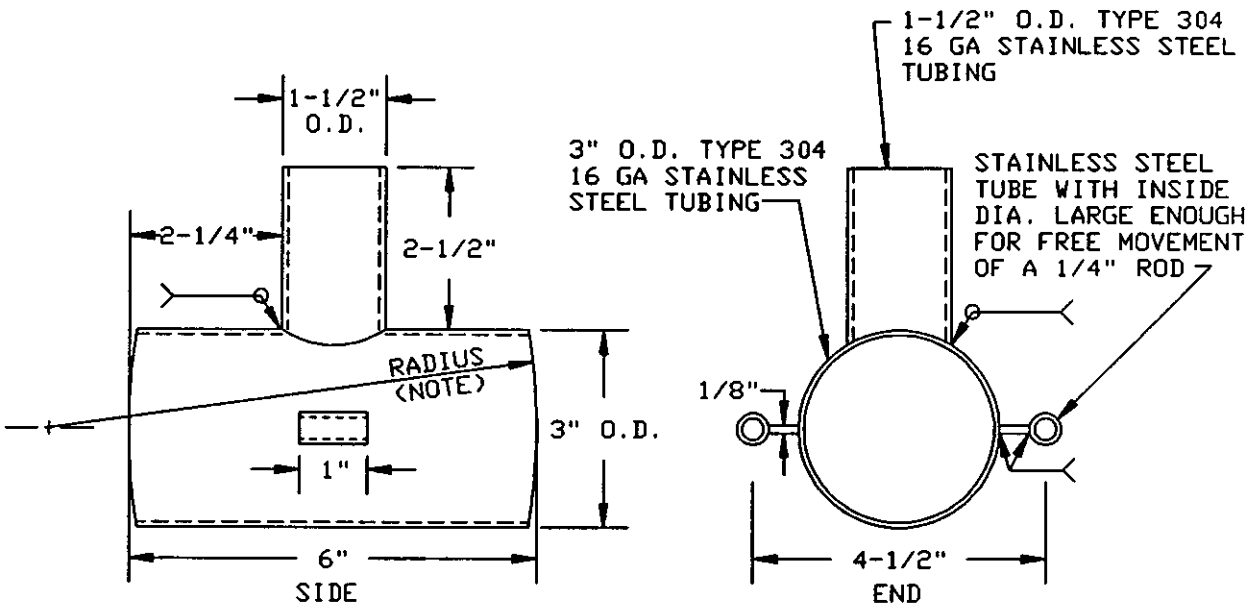


FIGURE 1B. Cylinder assembly. (1 required)

Note:

1. The radius shall be the same as the inside radius of the tank from which specimens are cut.

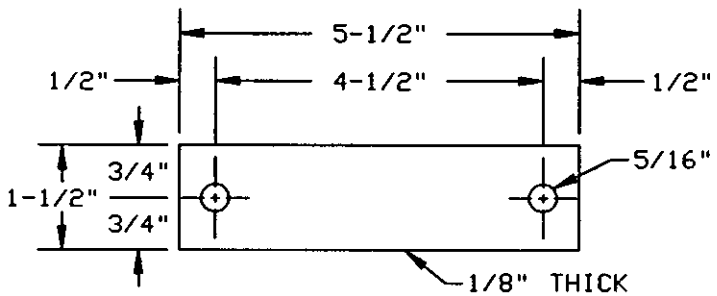


FIGURE 1C. Clamping strip. (2 required)

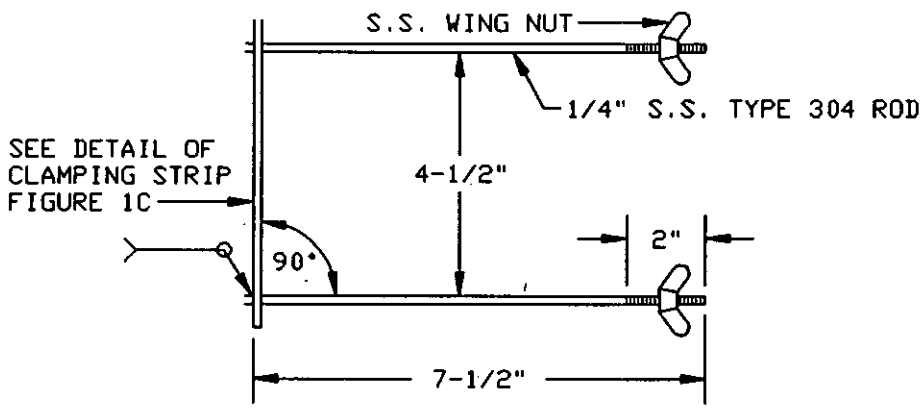


FIGURE 1D. Clamping assembly.

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	MIL-H-16451F	920410

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4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
DANNY C. MUI	805-982-5666	551-5666
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1562G) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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